DATA SET DS1509

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

1.5 7 4 -1.5 5.5 3 5.5

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

0.1 0.7 0.3 -0.1 0.5 0.2 0.5

a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.

b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

a) Wilcoxon sign-rank

H_0: 兩母体分佈相同

H_a: 〃 分佈不完全相同

$T^+ = 1.5 + 7 + \quad + 4 + 5.5 + 3 + 5.5 = 26.5$

$T^- = 1.5$

$\min(T^+, T^-) = 1.5$ 查表 夠小?

$n = 7$, $\alpha = 0.05$, 双尾 = 2

$1.5 < 2 \Rightarrow$ Reject H_0

b) R.R. $X \le 1$ or $X \ge 6$ p-value

(因有一個 $d < 0$)

$(C^7_0 + C^7_1 + C^7_6 + C^7_7) \times 0.5^7$

$= 0.125 > 0.05$

$\Rightarrow$ Non-reject H_0 不同於 (a)

DATA SET DS1511

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

Month	A	B		
1	3	7	-4	-6
2	14	12	2	3.5
3	7	9	-2	-3.5
4	10	15	-5	-7
5	9	12	-3	-5
6	6	6	0	
7	13	12	1	1.5
8	6	5	1	1.5
9	7	13	-6	-8

a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.

b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

11.

a) 成對比較 $\Rightarrow T^+$ or T^-

H_0: M_A and M_B 's breakdown rate 一樣

H_a: 〃 不一樣

$T^+ = 2 + X + 1 + 1 + 3.5 + 1.5 + 1.5 = 6.5$

$T^- = 29.5$

$\min(T^+, T^-) = 6.5$ 查表 $n = 8$, $\alpha = 0.05$ 双尾

$\Rightarrow$ critical = 4

$6.5 > 4 \Rightarrow$ Non-reject H_0

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

DATA SET DS1514

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5	11	17	8
2	24	9	12	20	16
3	20	5	13	20	10
4	18	9	14	16	12
5	22	6	15	24	7
6	19	11	16	22	9
7	20	8	17	25	21
8	19	11	18	21	14
9	17	7	19	19	12
10	21	9	20	23	13

a. What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?

b. Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.

c. Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

15.

a) $\overset{+}{d} = 15, 15, 15, 9, 16, 8, 12, 8, 10, 12$
$9, 4, 10, 4, 17, 13, 4, 7, 7, 10$

H_0：兩母體分佈無顯著差異

H_a：分佈有顯著差異

$T^+ = \frac{20(20+1)}{2} = 210$

$T^- = 0$

$\min(T^+, T^-) = 0$ 查表，$n = 20$，双尾 $\alpha = 0.05$

$\Rightarrow$ critical $= 52$

$0 < 52 \Rightarrow$ Reject H_0，兩母体極顯著差異

且 $X = 20$，

p-value $= P(X \le 0)$ or $P(X \ge 20)$

$= C^{20}_{0}\, 0.5^{20} + C^{20}_{20}\, 0.5^{20} = 2 \times 0.5^{20} < 0.05$

$\Rightarrow$ Reject H_0

Applying the Basics

6. Legos® The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

DATA SET DS1518

Training Period (hours)			
.5	1.0	1.5	2.0
8 13	9 16	4 2	4 2
14 20	7 10	6 7.5	7 10
9 16	5 5	7 10	5 5
12 19	8 13	8 13	5 5
10 18	9 16	6 7.5	4 2
T_1	T_2	T_3	T_4

Use the Kruskal–Wallis H-Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

6.

Kruskal-Wallis H-Test

H_0：訓練和組裝時間分佈一樣

H_a：不一樣

先排 Rank

$$H = \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(N+1)$$

$$= \frac{12}{20 \times 21}\left(\frac{7396}{5} + \frac{3600}{5} + \frac{1600}{5} + \frac{576}{5}\right) - 3(20+1)$$

$= 12.27 \sim \chi^2_{k-1, 0.01} = \chi^2_{3, 0.01} = 11.345$

$12.27 > 11.345 \Rightarrow$ Reject H_0

DATA SET DS1520

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10–19	20–39	40–59	60–69
	29 21	24 8	37 38	28 18
	33 29.5	27 15	25 10.5	29 21
	26 12.5	33 29.5	22 5.5	34 33
	27 15	31 24	33 29.5	36 36.5
	39 39	21 3	28 18	21 3
	35 35	28 18	26 12.5	20 1
	33 29.5	24 8	30 23	25 10.5
	29 21	34 33	34 33	24 8
	36 36.5	21 3	27 15	33 29.5
	22 5.5	32 25.5	33 29.5	32 25.5
Total	309	275	295	282
	247.5	168	216.5	188

a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal–Wallis H-test with $\alpha = .01$.

b. Find the approximate p-value for the test in part a.

c. Since the F-test in Chapter 11 and the H-test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F-test produced the test statistic $F = 0.87$ with p-value $= 0.468$, compare this p-value with the p-value in part b and explain the implications of the comparison.

8.

a) $H_0: P_A = P_B = P_C = P_D$

$H_a: \exists P_i \neq P_j,\ i \neq j$

$$H = \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(N+1)$$

$$= \frac{12}{40 \times 41} \times \left(\frac{247.5^2}{10} + \frac{168^2}{10} + \frac{216.5^2}{10} + \frac{188^2}{10} \right) - 3(41)$$

$= 2.632 \sim$ 查表 $\chi^2_{k-1,0.01}$ 夠大?

$\Rightarrow \chi^2_{3,0.01} = 11.34$, $2.632 < 11.34 \Rightarrow$ Non-reject H_0

$\Rightarrow$ 沒有足夠證據說明四母体分佈有顯著差異

b) p-value = $df = 3$, 查表 $\chi^2_{3,}$ —

$= \chi^2_{3,0.1} = 6.25$, $\chi^2_{3,0.9} = 0.58$

$2.632 < 6.25 \Rightarrow$ p-value $> 0.1 \Rightarrow$ Non-reject H_0

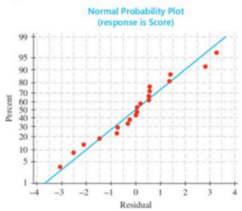
DATA SET DS1528

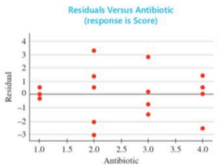
10. Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.[7] The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

	Antibiotic			
Child	1	2	3	4
1	4.8 2	2.2 1	6.8 4	6.2 3
2	8.1 2	9.2 3	6.6 1	9.6 4
3	5.0 3	2.6 1	3.6 2	6.5 4
4	7.9 2	9.4 4	5.3 1	8.5 3
5	3.9 3	7.4 4	2.1 2	2.0 1
	12	13	10	15

a. What design is used in collecting these data?

b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?





c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

10. 同一個人重複做測驗 $\Rightarrow F_r$

a) Random Block design

b) 常態性假設滿足

變異數同質性不符合

c) H_0: 四種測試分佈相同

H_a: 不完全相同

$$F_r = \frac{12}{bk(k+1)} \times \sum T_i^2 - 3b(k+1)$$

$$= \frac{12}{5 \times 4 \times 5} \times (12^2 + 13^2 + 10^2 + 15^2) - 3 \times 5(4+1)$$

$= 1.56 \sim \chi^2_{(k-1),0.05} = \chi^2_{3,0.05} = 7.815$

$1.56 < 7.815 \Rightarrow$ Non-reject H_0

DATA SET DS1529

11. Yield of Wheat Exercise 9 (Chapter 11 Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

	Location					
Variety	1	2	3	4	5	6
A	35.3 4	31.0 2	32.7 3	36.8 3	37.2 3	33.1 3
B	30.7 1	32.2 3	31.4 2	31.7 2	35.0 2	32.7 2
C	38.2 5	33.4 4	33.6 4	37.1 4	37.3 4	38.2 5
D	34.9 3	36.1 5	35.2 5	38.3 5	40.2 5	36.0 4
E	32.4 2	28.9 1	29.2 1	30.7 1	33.9 1	32.1 1

a. Use the Friedman F_r-test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.

b. When the analysis of variance was performed in Chapter 11, the ANOVA F-test produced the test statistic $F = 18.61$ with p-value $= .000$. How do these results compare with results of the Friedman F_r-test in part a? Explain.

11.

5種 wheat 在 6個地點 ⇒ F_r ⇒ 各自 Rank

a) H_0: 5 母体分佈相同

H_a: 5 母体分佈不完全相同

$$F_r = \frac{12}{bk(k+1)} \sum T_i^2 - 3b(k+1)$$

$$= \frac{12}{6\times5(6)} (18^2 + 12^2 + 26^2 + 27^2 + 7^2) - 3\times6\times5$$

$$= 20.133 \sim \chi^2_{k-1,0.05} = \chi^2_{4,0.05} = 9.49$$

$20.133 > 9.49$ ⇒ Reject H_0

⇒ 5種小麥分佈有顯著 different

b) p-value $= 0.000 < 0.05$ ⇒ Reject H_0